



NYC Yellow Taxi Analysis Documentation

Lynx Use Case – Data Analyst

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1. Purpose of this document

The purpose of this document is to explain the thought process behind the main design and analytical decisions taken throughout this assignment. It provides the rationale for the selected architecture, tools, data sources, and modeling approach, with the goal of making the work transparent and easy to follow.

This document is intended to complement the main analysis by showing not only **what** was done, but also **why** specific choices were made. It therefore serves as supporting documentation for the overall solution and helps illustrate the reasoning applied from data ingestion to final stakeholder communication.

2. Architecture

The architecture was designed to support the core activities of a Data Analyst in a robust and practical way. It combines widely used tools in the analytics workflow: **Pandas** for data processing and transformation, **SQL** for querying and validation, **Power BI** for visualization, and **PowerPoint** for stakeholder communication. This toolset was selected because it reflects common industry practice and allows the analysis to move efficiently from raw data to business-ready output.

The data model is built from two CSV sources, which are first processed in Pandas before being loaded into Power BI for visualization. The transformed dataset then acts as the foundation for the charts, while the visualizations are used to support the final presentation in PowerPoint. This creates a clear analytical flow from raw data to insights and then to executive communication.

In addition, both the raw and transformed versions of the data are retained. Keeping the original records intact is important for traceability, and the ability to reprocess the data if validation rules, business logic, or assumptions need to be updated later.

Below is a more detailed explanation of the rationale behind each tool used in the project.

- **Pandas:** was selected as the main data processing library because it is one of the most widely used Python tools for ETL workflows. Compared with a spreadsheet-based approach, a Pandas script provides full traceability, better reproducibility, and much greater flexibility for transformation and validation. In addition, Python is commonly used at Lynx, so including it in the project also helps demonstrate relevant technical skills.

- SQL: was used alongside Pandas because it is the standard language for querying and interacting with structured data. Although transformations could have been completed entirely in Pandas, using SQL adds another layer of realism and reflects one of the core tools used in Lynx's environment.
- Power BI: was chosen for visualization and dashboard development because BI tools are relevant for the role and are commonly used in business-facing analytics work. While visualizations could also have been built with Python libraries such as Seaborn, Power BI is a better choice here because it reflects the kind of reporting and dashboarding expected in a data analytics context.
- PowerPoint: was used to package the final insights into a stakeholder-friendly format.

3. Data Sources and Modelling

a. Data Sources

The analysis was built using three main data sources:

- **Yellow Taxi Trips**

This fact table was provided as part of the assignment and contains the raw NYC taxi trip records from 2018 through January 2023. Each row represents a single trip.

Source: https://github.com/m-gauer/NYC_TLC_Yellow_Taxi_Data

- **Data Dictionary Trip Records**

This document was also provided as part of the assignment and includes the glossary for all dataset fields. It was especially useful for interpreting coded variables and understanding the meaning of each attribute.

Source:

https://www.nyc.gov/assets/tlc/downloads/pdf/data_dictionary_trip_records_yellow.pdf

- **NYC TLC Taxi Zone Lookup**

This dimension table was referenced in the data dictionary and provides the Zone and Borough associated with each Area Code used in the fact table. Each row represents a unique taxi zone.

Source: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

b. Data Model

Because the analysis was based on a single fact table and did not require a more complex dimensional structure, the One Big Table (OBT) approach was the most appropriate choice. It was sufficient for the scope of this project and allowed the data to be analyzed in a simple and efficient way without adding unnecessary modeling complexity.

If the dataset had been more complex or had required multiple business processes and relationships between entities, a different approach such as a Kimball Dimensional Model would have been more suitable. In this case, however, OBT provided the right balance between simplicity, usability, and analytical flexibility.